Factory Primer

Everything you need before starting Track 1 or Track 2

> Three minutes. No manufacturing background required.

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1. Quick response, in three paragraphs

Twenty years ago a clothing brand ordered 100,000 units nine months before the season and hoped. Zara proved you could instead put a small batch in stores, watch what sells for a week, and reorder the winners within a fortnight. Shein industrialised the idea, and most of the industry now runs some version of it.

For the factory this changes the shape of the work completely: instead of a dozen huge orders planned months ahead, there are **hundreds of small orders in the building at once**, and the most profitable ones — the reorders of whatever is selling — **arrive without warning and are due in one to two weeks**.

Two consequences drive both tracks. First, nobody can hold the whole picture in their head any more, so the general manager spends the day asking *"what is actually happening right now — and can we take this new order?"* — building their co-pilot is **Track 1**. Second, the factory does not own enough capacity for its peaks, so batches are constantly farmed out to outside workshops, and *"who should make this?"* has to be answered within the hour — that is **Track 2**.

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2. The factory

SweaterCo makes knitwear for mall brands. Every order follows the same four steps:

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KNITTING  →  ASSEMBLY  →  WASHING  →  PACKING
(machines     (joining      (wash &      (labels, boxes,
 make the      panels into    press)       final check)
 panels)       a garment)
```

That is all the manufacturing knowledge the project needs. The factory works Monday to Saturday, and every order carries a customer, a piece count, a due date, and its current stage.

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3. The outside workshops (Track 2)

When the factory is full, batches are sent to one of **eight outside workshops** in the surrounding region. They differ in ways that matter: one is fast but sloppy, one is immaculate but tiny, one is cheap but slow. Each

has a profile card in `workshops.csv`, and before any batch can go anywhere, three questions must be answered: **can they make it** (not everyone is equipped for every product), **can they take it now** (capacity and current queue), and **are they allowed to** (one is suspended, one is on trial).

Today all three answers live in one dispatcher's head, and the choice is posted in a group chat with no record of the reasoning. Building the system that answers them instantly — and proving it allocates better than simple rules — is the whole of Track 2.

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4. The data

Three small, clean CSV files (see `data_dictionary.md` — half a page):

File	What it is
`orders.csv`	120 orders: who, what, how many, due when, at what stage
`workshops.csv`	Profile cards for the 8 outside workshops: what they make, capacity, queue, quality, price, status
`dispatch_requests.txt`	The group chat itself: 30 allocation requests as people typed them (Track 2's inbox)

The data is intentionally simple — you can read all of it in a spreadsheet in five minutes. The hard part of these tracks is the agent and its interface, not the data.

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5. Glossary

Term	Meaning
Quick response	Small first runs, then fast reorders of whatever sells
Reorder	A repeat of a style that is selling — urgent and unplanned
Order	One job in the factory
Stage	One of the four steps: knitting, assembly, washing, packing
Batch	An order (or part of one) sent to a workshop as one unit of work
Workshop / subcontractor	An outside shop that takes batches when the factory is full